Kaung Htet San

Cloud Engineer | Kubernetes | AWS | Terraform | Devops
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Career Profile

Cloud & Infrastructure Engineer with 8+ years of experience across cloud-native and hybrid environments. Currently working at NCS Singapore, focusing on Azure infrastructure using Terraform (IaC) to design, deploy, and manage scalable cloud environments.
Strong expertise in Terraform, including building reusable modules, resolving infrastructure drift, and managing enterprise Azure resources. Hands-on experience across cloud and infrastructure environments, managing system operations, Windows patching, capacity planning, and performance optimization. Skilled in Azure, AWS, Kubernetes, Linux, and OpenStack, with a solid foundation in both public and private cloud infrastructure. Focused on automation, reliability, and continuous improvement.

Technical Skills

- Cloud Platforms: Microsoft Azure & Amazon Web Services
- Infrastructure as Code (IaC) & Automation: Terraform
- Container & Orchestration: Kubernetes, Docker, Helm
- DevOps & CI/CD: Azure Devops, GitHub Actions, GitLab
- System Administration: Linux (RHEL, CentOS, Ubuntu), Windows Server (Patching), Shell scripting
- Monitoring & Security: Prometheus, Grafana, Zabbix, CloudWatch, Rsyslog, Vault
- Networking: Azure & AWS Networking (VNet/VPC, Private Endpoints), VPN, SSL/TLS, IAM, Security Groups, NACLs, DNS, AD, DHCP, TCP/IP, OSI Layers, NGINX

Certifications

- AWS Certified Solutions Architect - Associate (2024)
- Certified Kubernetes Administrator (CKA) (2024)
- Certified Kubernetes Application Developer (CKAD) (2025)
- Red Hat Certified System Administrator (RHCSA) (2023)
- VMware Certified Professional - Network Virtualization & Data Center Virtualization (2018)
- Microsoft Azure Administrator (2019) | Palo Alto Accredited Configuration Engineer (2019)
- Cisco Certified Network Associate (CCNA) (2018)

PROFESSIONAL EXPERIENCE

Cloud Engineer
NCS Singapore **November 2025 – Present**
Project : Inland Revenue Authority of Singapore (IRAS)

Responsibilities

- Architect and implement Azure infrastructure using Terraform (IaC) for enterprise environments, ensuring scalability, security, and maintainability.
- Design and enhance reusable Terraform modules to improve deployment consistency and align with infrastructure standards.
- Lead drift detection and remediation strategies, aligning Terraform state with live Azure environments and preventing configuration inconsistencies.
- Manage and optimize Azure services including Virtual Machines, networking, Private Endpoints, and Dev/Test environments, ensuring production reliability.

- Windows Server patching and lifecycle management, maintaining compliance with organizational security standards.
- Drive capacity planning and performance analysis, translating resource metrics into actionable scaling and optimization decisions.
- Troubleshoot and resolve complex infrastructure and Terraform issues independently across multi-environment deployments.
- Work closely with cross-functional teams and stakeholders to improve infrastructure design, performance, and cost efficiency.
- Identify gaps in existing infrastructure and proactively implement improvements to enhance reliability and operational efficiency.

Achievements

- Led resolution of Terraform drift issues across multiple environments, improving infrastructure stability and deployment accuracy.
- Designed and implemented reusable Terraform modules, increasing deployment consistency and reducing manual effort.
- Delivered capacity planning insights that informed scaling decisions and improved resource utilization.

Senior Cloud Infrastructure Engineer
SunIVY Solutions Myanmar Co., Ltd **May 2023 - Jul 2025**
Project : ATOM Myanmar Project Nokia CBIS 18 and CBIS 20

Responsibilities

- Led cloud infrastructure operations for telecom platforms, supporting large-scale production environments.
- Designed and maintained high-availability architectures using AWS (EC2, VPC, S3, IAM) and OpenStack private cloud.
- Automated infrastructure provisioning using Terraform and Bash across AWS and OpenStack environments.
- Built internal environments to simulate disaster recovery and validate high-availability scenarios.
- Implemented monitoring solutions using Prometheus and Grafana with custom metrics and alerting.
- Strengthened system security and logging through Linux hardening and centralized rsyslog.
- Managed VM lifecycle operations, including provisioning, patching, and Ceph storage management.

Achievements

- Improved operational readiness for cloud migration by prototyping AWS-based architectures.
- Reduced incident troubleshooting time by ~30% through proactive monitoring and observability improvements.
- Led failure recovery scenarios across availability zones, minimizing SLA impact.
- Developed SOPs and documentation to improve team efficiency and incident response.

SunIVY Solutions Myanmar Co., Ltd
L2/L3 Cloud Operation Engineer **December 2019 - May 2023**
Project : Telenor Myanmar Project Contracted by Nokia - Nokia CBIS 17.5 and CBIS 18

Responsibilities

- Managed OpenStack-based telecom infrastructure, ensuring high availability through proactive patching and VM lifecycle management.
- Introduced Terraform in internal environments to prototype Infrastructure as Code adoption.
- Built containerized test environments using Docker and Linux scripting for pre-deployment validation.
- Performed system hardening, backup optimization, and supported network configurations (VPN, NAT, routing).

Achievements

- Enabled hybrid cloud experimentation by provisioning AWS (EC2, S3, IAM) for internal PoC environments.
- Maintained 99.9% uptime for telecom workloads under strict SLA requirements.
- Reduced deployment time by ~40% through automation of provisioning and patching processes.
- Improved monitoring and incident response using Zabbix alerting.

SunIVY Solutions Myanmar Co., Ltd
System Engineer **November 2018 - December 2019**
SunIVY Solutions Myanmar

- Migrated DNS and AD (LDAP/RBAC) services during data center upgrades.
- Configured Palo Alto firewalls for hybrid secure connectivity.
- Managed Windows and VMware infrastructure during a major migration project.

ICT STAR GROUP MYANMAR Company Limited (ISGM)
Junior System/Network Engineer **January 2018 - October 2018**

- Managed Windows systems, deployed VMware servers, and configured networks (LAN, WAN, WiFi, Access Points).
- Provided technical support and troubleshooting

Cloud & DevOps Project Highlights

- **Cloud Resume (Serverless)**
  - AWS S3, CloudFront, Route 53 with CI/CD (GitHub Actions)
  - Terraform-managed infra with CloudWatch + Lambda (Python)
  - GitHub: https://github.com/KHS-cpu/portfolio
- **EKS 2048 Game Deployment**
  - Deployed app on EKS using Terraform + Helm
  - Used LoadBalancer and Fargate
  - GitHub: https://github.com/KHS-cpu/eks-2048
- **BookInfo Microservices with ECS + Harbor + TLS**
  - Deployed containerized app on ECS/VM with Harbor (TLS)
  - Implemented RBAC and namespace isolation
  - GitHub: https://github.com/KHS-cpu/bookinfo

Educational Background

B.E (Mechanical Precision & Automation)
University of Technology Yatanarpon Cyber City (UT-YCC) **Dec 2010 - Jan 2017**

Executive Diploma in Network and Server Infrastructure
Euroskills Technical and Vocational Academy (In Association with Lincoln University) **2019**

Languages & Soft Skills

- Languages: English (Fluent), Burmese (Native), Mandarin Chinese (HSK 2)
- Soft Skills: Problem Solving, Team Collaboration, Incident Handling, Knowledge Sharing, Mentoring

ADDITIONAL DETAILS

- Currently based in Singapore
- References: Available upon request.